

<i>Unit Costs</i>	A Product(2/3 NM)			
<i>Fab Location(Cities)</i>	Silicon	Aus	Shz	Seo
Hsinchu	20002073.6	20002406.4	20000793.6	20001126.4
Tainan	20002073.6	20002406.4	20000793.6	20001126.4
Kaohsiung	20002073.6	20002406.4	20000793.6	20001126.4
Arizona	26000793.6	26000640	26002252.8	26002073.6
Taichung	0	0	0	0
Japan	0	0	0	0
TOTAL Demand	425	425	1275	1487.5

<i>Decision Point</i>	A Product(2/3 NM)			
<i>Fab Location(Cities)</i>	Silicon	Aus	Shz	Seo
Hsinchu	0	0	0	0
Tainan	0	0	0	0
Kaohsiung	0	0	0	0
Arizona	0	0	0	0
Taichung	0	0	0	0
Japan	0	0	0	0

<i>Constrains</i>	A Product(2/3 NM)			
<i>Fab Location(Cities)</i>	Silicon	Aus	Shz	Seo
Hsinchu	0			
Tainan	0			
Kaohsiung	0			
Arizona	0			
Taichung	0			
Japan	0			
Excess Demand	425	425	1275	1487.5

Total Fixed Costs	Total Variable Cost	Penalty Cost
\$ 36,000,000,000.00	\$ 70,935,918,489.60	\$ 215,400,000,000.00

TSMC Fixed Location Decision With ABC Product

Transit Cost + Unit Cost per 1,000 Units (Based Upon ABC Product)

		B Product(5 NM)		
Mun	Tok	Silicon	Aus	Shz
20001920	20001126.4	16002073.6	16002406.4	16000793.6
20001920	20001126.4	16002073.6	16002406.4	16000793.6
20001920	20001126.4	0	0	0
26001920	26001920	20800793.6	20800640	20802252.8
0	0	16002073.6	16002406.4	16000793.6
0	0	0	0	0
425	212.5	612	612	1836

Cost Per Unit 1,000 (Based Upon ABC Product)

		B Product(5 NM)		
Mun	Tok	Silicon	Aus	Shz
0	0	0	0	1200
0	0	0	0	636
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0

Excess/Constraints Per Unit 1,000 (Based Upon ABC Product)

		B Product(5 NM)		
Mun	Tok	Silicon	Aus	Shz
425	212.5	612	612	0

Results (In Units Per 1,000 wafers)

Costs All together
\$ 322,335,918,489.60

Demand	Opened Capacity	Total Available Capacity
12580	5400	7920

Products (2/3nm, 5nm, and 7nm)

Products for TSMC)

			C Product(7 NM)				
Seo	Mun	Tok	Silicon	Aus	Shz	Seo	Mun
16001126.4	16001920	16001126.4	9002073.6	9002406.4	9000793.6	9001126.4	9001920
16001126.4	16001920	16001126.4	9002073.6	9002406.4	9000793.6	9001126.4	9001920
0	0	0	0	0	0	0	0
20802073.6	20801920	20801920	0	0	0	0	0
16001126.4	16001920	16001126.4	9002073.6	9002406.4	9000793.6	9001126.4	9001920
0	0	0	0	0	0	0	0
2142	612	306	221	221	663	773.5	221

ts for TSMC)

			C Product(7 NM)				
Seo	Mun	Tok	Silicon	Aus	Shz	Seo	Mun
0	0	0	0	0	0	0	0
264.5	0	0	221	221	663	773.5	221
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
1089.5	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

C Products for TSMC)

			C Product(7 NM)				
Seo	Mun	Tok	Silicon	Aus	Shz	Seo	Mun
788	612	306	0	0	0	0	0

Unmet Demand	
7180	

	Associated Costs Constraint	
Tok	Fixed Value	Constraint (Per 1,000 bathces)
9001126.4	120000000000	1200
9001126.4	120000000000	3000
0	452000000000	1440
0	650000000000	720
9001126.4	120000000000	1200
0	240000000000	360
110.5		

	Location Variable	SUM Units Used per (1,000)
Tok	1 or 0	
0	1	1200
0	1	3000
0	0	0
0	0	0
110.5	1	1200
0	0	0



Tok
0

Key

All Models Weight Roughly the same

EQ=.125KG*100000

1 Batch assumed Per 1,000 Wafer

Not all Demand meant Due to Capacity

ABC Products Make up or 74% of Revenue

 Means not able to be made at Site

Products 2/3nm, 5 nm, and 7nm

\$ 30,000,000.00 Penalty Cost for Unmet Demand

Baseline Information

Transit Rate Tables - Per Site (Saudi Market + Freightos estimates)

Batch Size 1000
 Unit weight (kg/wafer): \$ 0.32 According to resource
 Batch weight (kg per 100,000): 320
 Cost per 1,000 wafers = Rate(B5) x Batch Weight(B6). Batch = 1,000 wafers = 40 FOUPs = 320 kg (8 kg/load)

Baseline Assumptions

Country Premium

Country	USA	JPN
Rates	1.30	1.2

Cost Assumptions Per Wafer

	A Cost	B Cost	C Cost
\$	20,000.00	\$ 16,000.00	\$ 9,000.00

Demand Forecast

DEMAND (300mm wafers / year)
 Global total wafers (TSMC Website): 17000000 ABC Total
 A=below3nm, B=5nm, C=7nm 0.25

Demand Forecast Allocation

Per City Allocation (W/Ratio)			33.78%
City	Share	Product A	
Silicon Valley	0.1	425000	
Austin	0.1	425000	
Shenzhen	0.3	1275000	
Seoul	0.35	1487500	
Munich	0.1	425000	
Tokyo	0.05	212500	
TOTAL	1	4250000	

N

HSINCHU (Taiwan) - makes A, B, C

Shipping

Destination	Rate (\$/kg)	Cost per 1,000
Silicon Valley	\$ 6.48	\$ 2,073.60
Austin	\$ 7.52	\$ 2,406.40
Shenzhen	\$ 2.48	\$ 793.60
Seoul	\$ 3.52	\$ 1,126.40
Munich	\$ 6.00	\$ 1,920.00
Tokyo	\$ 3.52	\$ 1,126.40

TAINAN (Taiwan) - makes A, B, C

Destination	Rate (\$/kg)	Cost per 1,000
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Silicon Valley	\$	6.48	\$	2,073.60
Austin	\$	7.52	\$	2,406.40
Shenzhen	\$	2.48	\$	793.60
Seoul	\$	3.52	\$	1,126.40
Munich	\$	6.00	\$	1,920.00
Tokyo	\$	3.52	\$	1,126.40

KAOHSIUNG (Taiwan) - makes A

Destination	Rate (\$/kg)		Cost per 1,000	
Silicon Valley	\$	6.48	\$	2,073.60
Austin	\$	7.52	\$	2,406.40
Shenzhen	\$	2.48	\$	793.60
Seoul	\$	3.52	\$	1,126.40
Munich	\$	6.00	\$	1,920.00
Tokyo	\$	3.52	\$	1,126.40

ARIZONA (US) - makes A, B

Destination	Rate (\$/kg)		Cost per 1,000	
Silicon Valley	\$	2.48	\$	793.60
Austin	\$	2.00	\$	640.00
Shenzhen	\$	7.04	\$	2,252.80
Seoul	\$	6.48	\$	2,073.60
Munich	\$	6.00	\$	1,920.00
Tokyo	\$	6.00	\$	1,920.00

Taichung - makes B,C

Destination	Rate (\$/kg)		Cost per 1,000	
Silicon Valley	\$	6.48	\$	2,073.60
Austin	\$	7.52	\$	2,406.40
Shenzhen	\$	2.48	\$	793.60
Seoul	\$	3.52	\$	1,126.40
Munich	\$	6.00	\$	1,920.00
Tokyo	\$	3.52	\$	1,126.40

JAPAN - makes A, B, C- Note that this cannot be made right now but it is important to know that this is a good

Destination	Rate (\$/kg)		Cost per 1,000	
Silicon Valley	\$	6.48	\$	2,073.60
Austin	\$	7.52	\$	2,406.40
Shenzhen	\$	3.04	\$	972.80
Seoul	\$	2.48	\$	793.60
Munich	\$	6.00	\$	1,920.00
Tokyo	\$	1.52	\$	486.40

Links For Info and Resources

Revenue-> Demand

17 Million wafers

Wafer Price

Air Freight

Air Freight

<https://investor.tsmc.com/english/encrypt/file>

<https://www.tsmc.com/english/dedicatedFour>

<https://siliconanalysts.com/guide/semiconduc>

<https://suaidglobal.com/insights/air-freight-co>

<https://www.freightos.com/freight-resources/f>

s

led 25-wafer FOUP, per Wikipedia/Entegris/Semitracks). Est., not TSMC contract rates.- This is our Assumptions*

Taiwan Semi- ABC~74.5(Multiply by Ratio) 1.351351351

12580000

0.36 0.13 ~Note this is 74% of prduct mix

	48.65%	17.57%
Product B	Product C	
	612000	221000
	612000	221000
	1836000	663000
	2142000	773500
	612000	221000
	306000	110500
	6120000	2210000
	12580000	Total Demand

ode Application

Product Cost to Make				Total Costs	
	A	B	C	Total Variable A	Total Variable B
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$ 20,002,073.60	\$ 16,002,073.60
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$ 20,002,406.40	\$ 16,002,406.40
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$ 20,000,793.60	\$ 16,000,793.60
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$ 20,001,126.40	\$ 16,001,126.40
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$ 20,001,920.00	\$ 16,001,920.00
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$ 20,001,126.40	\$ 16,001,126.40

\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$	20,002,073.60	\$	16,002,073.60
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$	20,002,406.40	\$	16,002,406.40
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$	20,000,793.60	\$	16,000,793.60
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$	20,001,126.40	\$	16,001,126.40
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$	20,001,920.00	\$	16,001,920.00
\$	20,000,000.00	\$ 16,000,000.00	\$ 9,000,000.00	\$	20,001,126.40	\$	16,001,126.40

\$	20,000,000.00		\$	20,002,073.60
\$	20,000,000.00		\$	20,002,406.40
\$	20,000,000.00		\$	20,000,793.60
\$	20,000,000.00		\$	20,001,126.40
\$	20,000,000.00		\$	20,001,920.00
\$	20,000,000.00		\$	20,001,126.40

\$	26,000,000.00	\$ 20,800,000.00		\$	26,000,793.60	\$	20,800,793.60
\$	26,000,000.00	\$ 20,800,000.00		\$	26,000,640.00	\$	20,800,640.00
\$	26,000,000.00	\$ 20,800,000.00		\$	26,002,252.80	\$	20,802,252.80
\$	26,000,000.00	\$ 20,800,000.00		\$	26,002,073.60	\$	20,802,073.60
\$	26,000,000.00	\$ 20,800,000.00		\$	26,001,920.00	\$	20,801,920.00
\$	26,000,000.00	\$ 20,800,000.00		\$	26,001,920.00	\$	20,801,920.00

	\$ 16,000,000.00	\$ 9,000,000.00		\$	16,002,073.60
	\$ 16,000,000.00	\$ 9,000,000.00		\$	16,002,406.40
	\$ 16,000,000.00	\$ 9,000,000.00		\$	16,000,793.60
	\$ 16,000,000.00	\$ 9,000,000.00		\$	16,001,126.40
	\$ 16,000,000.00	\$ 9,000,000.00		\$	16,001,920.00
	\$ 16,000,000.00	\$ 9,000,000.00		\$	16,001,126.40

1 alternative for the future

\$	24,000,000.00	\$ 19,200,000.00	\$ 10,800,000.00	\$	24,002,073.60	\$	19,202,073.60
\$	24,000,000.00	\$ 19,200,000.00	\$ 10,800,000.00	\$	24,002,406.40	\$	19,202,406.40
\$	24,000,000.00	\$ 19,200,000.00	\$ 10,800,000.00	\$	24,000,972.80	\$	19,200,972.80
\$	24,000,000.00	\$ 19,200,000.00	\$ 10,800,000.00	\$	24,000,793.60	\$	19,200,793.60
\$	24,000,000.00	\$ 19,200,000.00	\$ 10,800,000.00	\$	24,001,920.00	\$	19,201,920.00
\$	24,000,000.00	\$ 19,200,000.00	\$ 10,800,000.00	\$	24,000,486.40	\$	19,200,486.40

Transport Matrix	Product(2/3 NM)			
<i>Fab Location(Cities)</i>	Silicon	Aus	Shz	Seo
Hsinchu	2073.6	2406.4	793.6	1126.4
Tainan	2073.6	2406.4	793.6	1126.4
Kaohsiung	2073.6	2406.4	793.6	1126.4
Arizona	793.6	640	2252.8	2073.6
Taichung				
Japan				
TOTAL Demand	425	425	1275	1487.5

Product Cost Matrix	A Product(2/3 NM)			
<i>Fab Location(Cities)</i>	Silicon	Aus	Shz	Seo
Hsinchu	20000000	20000000	20000000	20000000
Tainan	20000000	20000000	20000000	20000000
Kaohsiung	20000000	20000000	20000000	20000000
Arizona	26000000	26000000	26000000	26000000
Taichung				
Japan				

Total Variable C

\$ 9,002,073.60
 \$ 9,002,406.40
 \$ 9,000,793.60
 \$ 9,001,126.40
 \$ 9,001,920.00
 \$ 9,001,126.40

\$ 9,002,073.60
\$ 9,002,406.40
\$ 9,000,793.60
\$ 9,001,126.40
\$ 9,001,920.00
\$ 9,001,126.40

\$ 9,002,073.60
\$ 9,002,406.40
\$ 9,000,793.60
\$ 9,001,126.40
\$ 9,001,920.00
\$ 9,001,126.40

\$ 10,802,073.60
\$ 10,802,406.40
\$ 10,800,972.80
\$ 10,800,793.60
\$ 10,801,920.00
\$ 10,800,486.40

Node Matrix Application

B Product(5 NM)						C Product(7 NM)			
Mun	Tok	Silicon	Aus	Shz	Seo	Mun	Tok	Silicon	Aus
1920	1126.4	2073.6	2406.4	793.6	1126.4	1920	1126.4	2073.6	2406.4
1920	1126.4	2073.6	2406.4	793.6	1126.4	1920	1126.4	2073.6	2406.4
1920	1126.4								
1920	1920	793.6	640	2252.8	2073.6	1920	1920		
		2073.6	2406.4	793.6	1126.4	1920	1126.4	2073.6	2406.4
425	212.5	612	612	1836	2142	612	306	221	221

[illegible]

				Associated Costs Constraint
Shz	Seo	Mun	Tok	Fixed Value (Billions)
793.6	1126.4	1920	1126.4	12
793.6	1126.4	1920	1126.4	12
				9.49
				65
793.6	1126.4	1920	1126.4	2.87
				24
663	773.5	221	110.5	

C Product(7 NM)			
Shz	Seo	Mun	Tok
9000000	9000000	9000000	9000000
9000000	9000000	9000000	9000000
9000000	9000000	9000000	9000000

Constraint (Millions)
1.2
3
1.44
0.72
1.2
1.2